Kai Ren

Kyoto University | Ph.D. Candidate | Intelligent Science and Technology
EMG Sensing | Robotics | Human-Computer Interaction
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Profile

Ph.D. candidate in Intelligent Science and Technology at Kyoto University, focusing on electromyography (EMG) sensing, deep learning, robotics, and human-computer interaction. Current work targets physiological signal modeling and motion intent prediction for seat-based and wearable assistive systems, with end-to-end experience from data acquisition to real-time system integration.

Research Experience

RIKEN, Man-Machine Collaboration Research Team, Japan Dec 2021 – Present
Research Trainee RIKEN-GRP Homepage

- Conduct EMG-based analysis and prediction for sit-to-stand and stand-to-sit transitions in assistive scenarios.
- Design real-time physiological signal processing pipelines and neural network models for motion intent prediction.
- Build experimental closed-loop systems for powered assistive devices, including seat-based prototypes.
- Execute the full research workflow: EMG hardware setup, data collection, model training, online evaluation, and validation.

Education

- Kyoto University, Japan, Ph.D. in Intelligent Science and Technology Apr 2021 – Present
- Ocean University of China, China, M.Eng. in Computer Application Engineering 2017 – 2020
- Ocean University of China, China, B.Eng. in Computer Science and Technology 2013 – 2017

Activities and Projects

- 2014 ACM-ICPC Shandong Provincial Contest participant.
- 2016 Project Lead, SRDP Program, Ocean University of China, iOS app development.
- 2025 IST Best Poster Award, Graduate School of Informatics, Kyoto University.
- Third-Class Academic Scholarship, 2017–2020, awarded consecutively each year.

Publications

1. Practical Motion Classification and Forecast toward Safe Assists of Sit-to-Stand and Stand-to-Sit Motions. Submitted to *IEEE ICRA 2027*; **under review**.
2. Anticipatory prediction of sit-to-stand and stand-to-sit transitions: a unified approach. *Frontiers in Bioengineering and Biotechnology*, 2026. DOI: 10.3389/fbioe.2026.1792582.
3. Integrated Motion State Prediction for Sit-to-Stand and Stand-to-Sit Motions Toward Effective Power Assist Control. *IEEE ICRA*, 2025, pp. 947–952.
4. Q-DFN: Q-learning Decisive Fusion Network for Object Detection. *Proceedings of CSAI 2019*, pp. 88–92.

Technical Skills

- **Programming:** Python, MATLAB, C++
- **Machine Learning:** PyTorch, deep learning for biosignals and time-series modeling
- **Signal Processing:** EMG acquisition, preprocessing, feature extraction, real-time processing
- **Systems and Hardware:** real-time inference, closed-loop systems, assistive-device integration, EMG sensor setup
- **Tools:** Linux, Git, experimental system integration

Research Interests and Languages

Research Interests: physiological signal modeling for motion intent prediction, multimodal perception, real-time reasoning, and human-centered AI.
Languages: Chinese, native; English, TOEFL; Japanese, JLPT N2.